Example 6.13

Find the volume of the parallelepiped whose coterminus edges are given by the vectors $2\hat{i}-3\hat{j}+4\hat{k}, \hat{i}+2\hat{j}-\hat{k}$ and $3\hat{i}-\hat{j}+2\hat{k}$.

Solution

We know that the volume of the parallelepiped whose coterminus edges are $\vec{a},\vec{b},\vec{c}$ is given by $|[\vec{a},\vec{b},\vec{c}]|$. Here, $\vec{a}=2\hat{i}-3\hat{j}+4\hat{k}$, $\vec{b}=\hat{i}+2\hat{j}-\hat{k}$, $\vec{c}=3\hat{i}-\hat{j}+2\hat{k}$.

Since $[\vec{a},\vec{b},\vec{c}]=\begin{vmatrix}2 & -3 & 4\\ 1 & 2 & -1\\ 3 & -1 & 2\end{vmatrix}=-7$, the volume of the parallelepiped is $|-7|=7$ cubic units.

Example 6.14

Show that the vectors $\hat{i}+2\hat{j}-3\hat{k}$, $2\hat{i}-\hat{j}+2\hat{k}$ and $3\hat{i}+\hat{j}-\hat{k}$ are coplanar.

Solution

Here, $\vec{a}=\hat{i}+2\hat{j}-3\hat{k}$, $\vec{b}=2\hat{i}-\hat{j}+2\hat{k}$, $\vec{c}=3\hat{i}+\hat{j}-\hat{k}$

We know that $\vec{a},\vec{b},\vec{c}$ are coplanar if and only if $[\vec{a},\vec{b},\vec{c}]=0$. Now, $[\vec{a},\vec{b},\vec{c}]=\begin{vmatrix}1 & 2 & -3\\ 2 & -1 & 2\\ 3 & 1 & -1\end{vmatrix}=0$.

Therefore, the three given vectors are coplanar.

Example 6.15

If $2\hat{i}-\hat{j}+3\hat{k}$, $3\hat{i}+2\hat{j}+\hat{k}$, $\hat{i}+m\hat{j}+4\hat{k}$ are coplanar, find the value of m.

Solution

Since the given three vectors are coplanar, we have $\begin{vmatrix}2 & -1 & 3\\ 3 & 2 & 1\\ 1 & m & 4\end{vmatrix}=0\Rightarrow m=-3$.

Example 6.16

Show that the four points $(6,-7,0)$, $(16,-19,-4)$, $(0,3,-6)$, $(2,-5,10)$ lie on a same plane.

Solution

Let $A=(6,-7,0)$, $B=(16,-19,-4)$, $C=(0,3,-6)$, $D=(2,-5,10)$. To show that the four points A,B,C,D lie on a plane, we have to prove that the three vectors $\overrightarrow{AB},\overrightarrow{AC},\overrightarrow{AD}$ are coplanar.

Now, $\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}=(16\hat{i}-19\hat{j}-4\hat{k})-(6\hat{i}-7\hat{j})=10\hat{i}-12\hat{j}-4\hat{k}$

$\overrightarrow{AC}=\overrightarrow{OC}-\overrightarrow{OA}=-6\hat{i}+10\hat{j}-6\hat{k}$ and $\overrightarrow{AD}=\overrightarrow{OD}-\overrightarrow{OA}=-4\hat{i}+2\hat{j}+10\hat{k}$.

We have $[\overrightarrow{AB},\overrightarrow{AC},\overrightarrow{AD}]=\begin{vmatrix}10 & -12 & -4\\ -6 & 10 & -6\\ -4 & 2 & 10\end{vmatrix}=0$.

Therefore, the three vectors $\overrightarrow{AB},\overrightarrow{AC},\overrightarrow{AD}$ are coplanar and hence the four points $A,B,C,$ and D lie on a plane.

Chapter 6 Vector Algebra.indd 236 31-01-2020 17:50:42